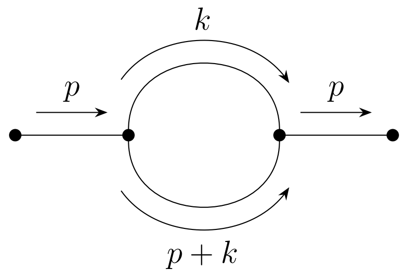
1. One loop in ϕ^3

Simply speaking, renormalization is to solve the problem of divergent when we try to solve the loop graphs by subtracting the divergent part in the expression. In the following, we will use ϕ^3-theory as an example to discuss the renormalization.

Firstly, the Lagrangian of ϕ^3-theory is:

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{g}{3!}\phi^3. \tag{1}$$

As the simplest case, the 1-loop correlation for the propagator is (notice that in this graph ,there are **all inner lines**, i.e., propagators):



where the momentum k is **off-shelled**, which means k doesn't follow the relation $k^2 + E_k^2 = m^2$, so technically k could be very large. But we don't need to worry about it, because k only appears in the loop, which means that it must follows the 4-momentum conservation. Thus we need to sum over all possible k values, which finally will contribute a integral measurement when we calculate the loop part. Thus, generally, we can write the whole graph as

$$\frac{\mathrm{i}}{p^2 - m^2 + \mathrm{i}\epsilon}\left[-\mathrm{i}\Sigma_1(p^2)\right]\frac{\mathrm{i}}{p^2 - m^2 + \mathrm{i}\epsilon}, \tag{2}$$

where $-\mathrm{i}\Sigma_1(p^2)$ represent the loop part, i.e.,

$$-\mathrm{i}\Sigma_1(p^2) = \frac{g^2}{2}\int\frac{\mathrm{d}^4k}{(2\pi)^4}\frac{\mathrm{i}}{k^2 - m^2 + \mathrm{i}\epsilon}\frac{\mathrm{i}}{(p+k)^2 - m^2 + \mathrm{i}\epsilon}, \tag{3}$$

where we will note the integral measurement $\frac{\mathrm{d}^4k}{(2\pi)^4} \to \mathrm{d}^4k$ for simple. Next, we will focus on this integral as an example to talk about how to solve this type of integral.

Wick Rotation

The first step is called *Wick Rotation* . The main idea of this procedure is transferring p^0 from Minkowski space into Euclid space. Starting from changing the variable $t = -\mathrm{i}\tau$ thus $k^0 = \mathrm{i}\omega$, so the measurement becomes

$$\mathrm{d}^4k = \mathrm{d}k^0\mathrm{d}^3\vec{k} = \mathrm{i}\mathrm{d}\omega\mathrm{d}^3\vec{k}. \tag{4}$$

In the p^0 prat of the integral

$$\int \mathrm{d}k^0\frac{\mathrm{i}}{(k^0)^2 - E_k^2}\frac{\mathrm{i}}{(p^0+k^0)^2 - E_{p+k}^2}, \tag{5}$$

poles are:

$$k^0 = \pm E_k, \quad k^0 = -p^0 \pm E_{p+k}. \tag{6}$$

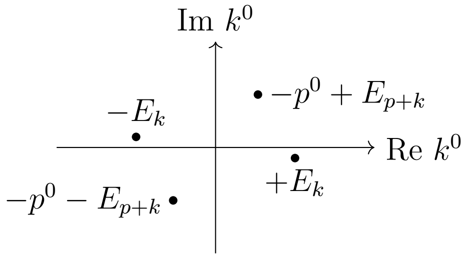
After Wick rotation:

$$\int \mathrm{i}\mathrm{d}\omega \frac{\mathrm{i}}{-\omega^2 - E_k^2} \frac{\mathrm{i}}{(p^0 + \mathrm{i}\omega)^2 - E_{p+k}^2}, \tag{7}$$

the poles become to:

$$p^0 = \mathrm{i}p_E^0: \quad \omega = \pm E_k - \mathrm{i}\epsilon, \quad \omega = -p_E^0 \pm E_{p+k}. \tag{8}$$

There are three cases for the poles. First, if p^0 is imaginary $\implies p^0 = \mathrm{i}p_E^0$, so we have



Another two case are for $p^0 = p_E^0$ is real, and if $|p^0| < m$, we have: $\omega = \pm E_k - \mathrm{i}\epsilon, \quad \omega = -p^0 \pm E_{p+k} - \mathrm{i}\epsilon$. If $|p^0| > m$, we have: $\omega = \pm E_k - \mathrm{i}\epsilon, \quad \omega = -p^0 \pm E_{p+k} - \mathrm{i}\epsilon$. We can see that no matter in which case, the Wick rotation always keep the same result with for the counter integral since the integration path including the same poles.

So the divergence in Eq.(3) is called the UV divergence, which can be seen by firstly taking the polar coordinate: $\mathrm{d}^4k_E = k_E^3\mathrm{d}k_E\mathrm{d}\Omega$, thus in large k

$$\Sigma_1 \sim \int \mathrm{d}^d k \frac{1}{k^4} \sim \int \mathrm{d}k\mathrm{d}\Omega \frac{1}{k} \sim \int \frac{\mathrm{d}k}{k} \sim \ln \Lambda \to \infty. \tag{9}$$

As we said above, renormalization is a way to cancel the divergent. So first of all, we need to recognize which part is finite and which part is divergent. Thus, we introduce a finite momentum μ to avoid new divergent in $k = 0$, so we have

$$\begin{aligned}\Sigma_1 = \frac{\mathrm{i}g^2}{32\pi^4} \Bigg[\int \mathrm{d}^4k \Bigg\{ &\frac{1}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)} - \frac{1}{(k^2 - \mu^2 + \mathrm{i}\epsilon)^2} \Bigg\} \\ &+ \int \mathrm{d}^4k \, \frac{1}{(k^2 - \mu^2 + \mathrm{i}\epsilon)^2} \Bigg].\end{aligned} \tag{10}$$

In a large k, the expression inside the first integral behaves as

$$\frac{1}{k^4}\left(1 + O\left(\frac{1}{k^2}\right)\right) - \frac{1}{k^4} = O\left(\frac{1}{k^6}\right). \tag{11}$$

Thus $\int \mathrm{d}^4k/k^6$ is UV finite, and the logarithmic UV divergence has been isolated in the second integral. This second integral is independent of the external momentum p, which is why the corresponding counterterm is local.

To understand the divergent intergal, we need to work in a space-time lattice with spacing a, the free propagator is then $S_F(k; \mu, a)$, so we have

$$\Sigma_1 = \frac{\mathrm{i}g^2}{32\pi^4} \left[\int \mathrm{d}^4\, k S_F(k; m, a) S_F(p + k; m, a) - S_F^2(k; \mu, a) + \int \mathrm{d}^4 k \, S_F^2(k; m, a) \right] \tag{12}$$

$$\equiv \Sigma_{1,\mathrm{fin}}(p, m, \mu, a) + \Sigma_{1,\mathrm{div}}(m, \mu, a). \tag{13}$$

The $\Sigma_{1,\text{div}}(m,\mu,a)$ in the lattice diverges as

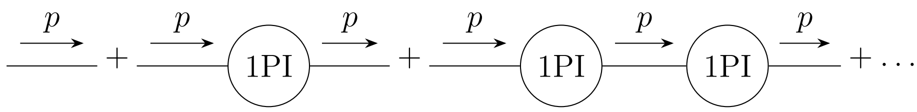
$$\Sigma_{1,\text{div}}(m,\mu,a) = \frac{\mathrm{i}g^2}{32\pi^4}\int \mathrm{d}^4k\, S_F^2(k;\mu,a) \tag{14}$$

$$= -\frac{\mathrm{i}g^2}{32\pi^4}\int_{\omega^2+\vec{k}^2<\frac{1}{a^2}} \mathrm{d}\omega \mathrm{d}^3\vec{k}\, \frac{1}{(\omega^2+\vec{k}^2+\mu^2)^2} \tag{15}$$

$$= -\frac{\mathrm{i}g^2}{16\pi^2}\int_0^{1/a} \mathrm{d}k\, \frac{k^3}{(k^2+\mu^2)^2} = -\frac{g^2}{16\pi^2}\ln\frac{1}{a}, \quad \text{as } a\to 0. \tag{16}$$

This result just as same as our analysis in Eq.(9).

Thus we can interpret the divergence in a more general way. We consider all the loop corrections for the propagator. Let $\mathrm{i}\Sigma$ be the sum of 1-PI, the propagator is then

$G_2(p^2) =$  (17)

$$= \frac{\mathrm{i}}{p^2-m^2} + \frac{\mathrm{i}}{p^2-m^2}(-\mathrm{i}\Sigma)\frac{\mathrm{i}}{p^2-m^2} + \frac{\mathrm{i}}{p^2-m^2}(-\mathrm{i}\Sigma)\frac{\mathrm{i}}{p^2-m^2}(-\mathrm{i}\Sigma)\frac{\mathrm{i}}{p^2-m^2} + \dots \tag{18}$$

$$= \frac{\mathrm{i}}{p^2-m^2} + \frac{\mathrm{i}}{p^2-m^2}\frac{\Sigma}{p^2-m^2} + \frac{\mathrm{i}}{p^2-m^2}\left(\frac{\Sigma}{p^2-m^2}\right)^2 + \dots \tag{19}$$

$$= \frac{\mathrm{i}}{p^2-m^2}\frac{p^2-m^2}{p^2-m^2-\Sigma} \tag{20}$$

$$= \frac{\mathrm{i}}{p^2-m^2-\Sigma} \equiv \frac{\mathrm{i}}{p^2-m_{\text{ph}}^2}, \tag{21}$$

where we have used $\sum_{i=0}^{\infty} a_i = \frac{1}{1-a}$. And $m_{\text{ph}}^2 = m^2 + \Sigma(p^2 = m_{\text{ph}}^2, m^2)$ in the pole define as $p^2 = m_{\text{ph}}^2$. The physical mass m_{ph} is the mass that can be directly observed. So the goal is changing Σ_1 to Σ_{1R}, which all the mass in it should be physical mass, i.e.,

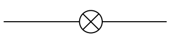
$$\Sigma_1 = \frac{\mathrm{i}g^2}{32\pi^4}\int \mathrm{d}^4k \frac{1}{(k^2-m^2+\mathrm{i}\epsilon)((p+k)^2-m^2+\mathrm{i}\epsilon)} \tag{22}$$

$$\Longrightarrow \Sigma_1 = \frac{\mathrm{i}g^2}{32\pi^4}\int \mathrm{d}^4k \frac{1}{(k^2-m_{\text{ph}}^2+\mathrm{i}\epsilon)((p+k)^2-m_{\text{ph}}^2+\mathrm{i}\epsilon)}. \tag{23}$$

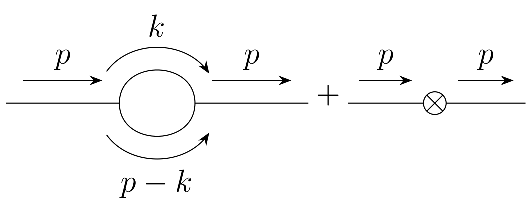
For that, we suppose the mass term in Lagrangian satisfies

$$\frac{1}{2}m_0^2\phi^2 = \frac{1}{2}m_{\text{ph}}^2\phi^2 + \frac{1}{2}\delta m^2\phi^2 \tag{24}$$

The last term is called the **counterterm**, which will contribute a graph of

 $= -\mathrm{i}\delta m^2.$

Thus, the renormalized self-energy to $O(g^2)$ is defined as

$-\mathrm{i}\Sigma_{1R} =$  $= -\mathrm{i}\Sigma_1 - \mathrm{i}\delta m^2$ (25)

$$\Longrightarrow \Sigma_{1R} = \Sigma_1 + \delta m^2 = (\Sigma_{1,\text{div}} + \delta m^2) + \Sigma_{1,\text{fin}} \tag{26}$$

Since the counterterm δm^2 can be chosen freely, we can then always choose $\delta m^2 = -\Sigma_{1,\mathrm{div}}$, such that the UV-divergent cancels.

Therefroe, to calculate Σ_{1R}, we have many ways, one way is firstly differentiate with respect to p^2, and use the fact that only Σ_1 contains p^μ, so we have

$$\frac{\partial \Sigma_{1R}}{\partial p^2} = \frac{\partial p^\mu}{\partial p^2}\frac{\partial}{\partial p^\mu}\Sigma_{1R} = \frac{p^\mu}{2p^2}\frac{\partial}{\partial p^\mu}\Sigma_1 \tag{27}$$

$$= \frac{\mathrm{i}g^2}{32\pi^4}\frac{p^\mu}{2p^2}\int \mathrm{d}^4k \frac{\partial}{\partial p^\mu}\left[\frac{1}{(k^2-m^2+\mathrm{i}\epsilon)((p+k)^2-m^2+\mathrm{i}\epsilon)} - \frac{1}{(k^2-\mu^2+\mathrm{i}\epsilon)^2}\right] \tag{28}$$

$$= \frac{\mathrm{i}g^2}{64\pi^4}\frac{p^\mu}{2p^2}\int \mathrm{d}^4k \frac{-2(p+k)_\mu}{(k^2-m^2+\mathrm{i}\epsilon)((p+k)^2-m^2+\mathrm{i}\epsilon)^2} \tag{29}$$

$$= \frac{-\mathrm{i}g^2}{32\pi^4 p^2}\int \mathrm{d}^4k \frac{p\cdot(p+k)}{(k^2-m^2+\mathrm{i}\epsilon)((p+k)^2-m^2+\mathrm{i}\epsilon)^2}. \tag{30}$$

To calculate this integral, we need to introduce a Feynman parameter x and use the fact:

$$\frac{1}{AB^n} = \int_0^1 \mathrm{d}x \frac{nx^{n-1}}{(xB+(1-x)A)^{n+1}}. \tag{31}$$

Thus,

$$\frac{\partial \Sigma_{1R}}{\partial p^2} = \frac{-\mathrm{i}g^2}{32\pi^4 p^2}\int_0^1 \mathrm{d}x \int \mathrm{d}^4k \frac{2p\cdot(p+k)x}{(-m_{\mathrm{ph}}^2 + p^2x + 2p\cdot kx + k^2 + \mathrm{i}\epsilon)^3} \tag{32}$$

$$= \frac{\mathrm{i}g^2}{16\pi^4 p^2}\int_0^1 \mathrm{d}x \int \mathrm{d}^4k \frac{p\cdot(p+k)x}{(m_{\mathrm{ph}}^2 - p^2x - 2p\cdot kx - k^2 - \mathrm{i}\epsilon)^3} \tag{33}$$

$$= \frac{\mathrm{i}g^2}{16\pi^4 p^2}\int_0^1 \mathrm{d}x \int \mathrm{d}^4k \frac{p\cdot(p+k+px-px)x}{(m_{\mathrm{ph}}^2 - p^2x - (k+px)^2 + p^2x^2 - \mathrm{i}\epsilon)^3}, \tag{34}$$

let $k' = k + px$

$$\frac{\partial \Sigma_{1R}}{\partial p^2} = \frac{\mathrm{i}g^2}{16\pi^4 p^2}\int_0^1 \mathrm{d}x \int \mathrm{d}^4k' \frac{p^2(1-x)x + p\cdot k'x}{(m_{\mathrm{ph}}^2 - p^2x(1-x) - k'^2 - \mathrm{i}\epsilon)^3} \tag{35}$$

$$= \frac{\mathrm{i}g^2}{16\pi^4 p^2}\int_0^1 \mathrm{d}x \int \mathrm{d}^4k' \frac{p^2x(1-x)}{(m_{\mathrm{ph}}^2 - p^2x(1-x) - k'^2)^3} \tag{36}$$

$$= \mathrm{i}g^2 \int_0^1 \mathrm{d}x\, x(1-x) \int \mathrm{d}^4k' \frac{1}{(k'^2-\Delta)^3}. \tag{37}$$

Use the integral fact

$$\int \frac{\mathrm{d}^d l}{(2\pi)^d}\frac{1}{(l^2-\Delta)^m} = \frac{\mathrm{i}(-1)^m}{(4\pi)^{d/2}}\frac{\Gamma(m-d/2)}{\Gamma(m)}\Delta^{d/2-m}, \tag{38}$$

with $\Delta = m_{\mathrm{ph}}^2 - p^2x(1-x)$. We then have

$$\frac{\partial \Sigma_{1R}}{\partial p^2} = -\frac{g^2}{32\pi^2}\int_0^1 \mathrm{d}x \frac{x(1-x)}{m_{\mathrm{ph}}^2 - p^2x(1-x)} \tag{39}$$

$$= \frac{g^2}{32\pi^2}\frac{\partial}{\partial p^2}\int_0^1 \mathrm{d}x \ln[m_{\mathrm{ph}}^2 - p^2x(1-x)]. \tag{40}$$

Thus, we can integrate p^2 in both side, then we read

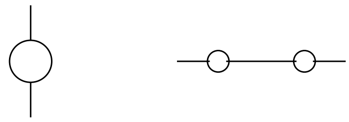
$$\Sigma_{1R} = \frac{g^2}{32\pi^2} \int_0^1 \mathrm{d}x \ln[m_{\text{ph}}^2 - p^2 x(1-x)] + C. \tag{41}$$

The boundary condition is at $p^2 = m_{\text{ph}}^2$, $\Sigma_{1R} = 0$, so we can determine the integration constant, therefore, we obtain the final result as:

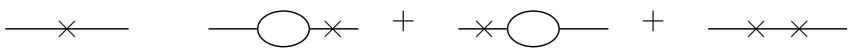
$$\Sigma_{1R} = \frac{g^2}{32\pi^2} \int_0^1 \mathrm{d}x \ln\left[\frac{m_{\text{ph}}^2 - p^2 x(1-x)}{m_{\text{ph}}^2(1-x+x^2)}\right]. \tag{42}$$

For higher order graphs, the same idea is used inside larger graphs. If a one-loop bubble appears as a subgraph, we renormalize that small part first:

1. First draw the graph before the subgraph is subtracted. The one-loop bubble may sit vertically on another line, or several bubbles may sit on the same propagator line:



2. Then replace one or more divergent bubbles by their counterterm. The cross means "use the local counterterm instead of the loop":



3. After adding the original graph and these counterterm graphs, the divergent bubble is replaced by its finite renormalized value $-\mathrm{i}\Sigma_{1R}$.

In words, we do not wait until the whole large graph is finished. We remove the divergence as soon as a divergent subgraph appears. This is why counterterm graphs have to be included together with ordinary loop graphs.

Degree of divergence

$$\Sigma_1(p^2, m^2, d) = \frac{\mathrm{i}g^2}{2(2\pi)^d} \int \mathrm{d}^d k \frac{1}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)} \text{ has } d-4 \text{ dim for } k \tag{43}$$

we call it as degree of divergence of the graph.

$$\text{If } d < 4 \implies \text{graph is convergent.} \tag{44}$$

$$d \geq 4 \implies \text{graph is divergent} \implies \text{needs to be renormalized} \tag{45}$$

Differentiating once

$$\frac{\partial}{\partial p^\mu}\Sigma_1 = \frac{-\mathrm{i}g^2}{(2\pi)^d} \int \mathrm{d}^d k \frac{(p+k)_\mu}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)^2} \tag{46}$$

degree of divergence is $d-5$. if $d=5$, it diverges logarithmically.

Symmetrizing by $k \to -k$ will kill the divergence. Differentiating again:

$$\frac{\partial^2}{\partial p^\mu \partial p^\nu}\Sigma_1 = \frac{\mathrm{i}g^2}{(2\pi)^d}\int \mathrm{d}^d k \frac{2(p+k)_\mu (p+k)_\nu - g_{\mu\nu}((p+k)^2 - m^2)}{(k^2-m^2)^2((p+k)^2-m^2)^3} \tag{47}$$

The degree of divergence is $d-6$. Integrating twice will get Σ_{1R} with 2 constants in the form: $A + B_\mu p^\mu$. Since Σ must be Lorentz invariant, thus $B_\mu = 0$. Thus only the mass term is the counterterm.

If $d = 6$, differentiating Σ_1 3 times will get finite results with degree of divergence $d-7$. Integrating gives the constant $A + Bp^2 + C_\mu p^\mu$, where $C_\mu = 0$ because of Lorentz invariance condition. Thus, the counterterm includes the mass and another proportion to p^2, i.e.

$$\delta m^2 - \delta Z p^2 \tag{48}$$

generated from the Lagrangian

$$-\frac{1}{2}\delta m^2 \phi^2 + \frac{1}{2}\delta Z(\partial \cdot \phi)^2 \subset \mathcal{L} \tag{49}$$

where δZ is an example of wave function renormalization. We can understand it from the bare propagator:

$$G_{2(0)} = \langle 0|\, \mathrm{T}\{\phi_0(p)\phi_0(0)\}\, |0\rangle \tag{50}$$

$$= Z\, \langle 0|\, \mathrm{T}\{\tilde{\phi}(p)\phi(0)\}\, |0\rangle = ZG_2 \quad \text{(renormalized propagator)} \tag{51}$$

For p^2 not at the pole ($p^2 = m_{ph}^2$), so the propagator cannot be written as

$$G_2(p^2 = m_{ph}^2) = \frac{\mathrm{i}}{p^2 - m_{ph}^2} \tag{52}$$

but only be

$$G_2(p^2) = \frac{\mathrm{i}}{p^2 - m_0^2 - \Sigma_1(p^2)} = \frac{\mathrm{i}}{p^2 - m_0^2 - \Sigma_1(m_{ph}^2) + \Sigma_1(m_{ph}^2) - \Sigma_1(p^2)} \tag{53}$$

$$\text{where } \Sigma_1(m_{ph}^2) = m_{ph}^2 - m_0^2, \quad \Sigma_{1R}(p^2) = \Sigma_1(p^2) - \Sigma_1(m_{ph}^2) \tag{54}$$

$$= \frac{\mathrm{i}}{p^2 - m_{ph}^2 - \Sigma_{1R} + \mathrm{i}\epsilon} \tag{55}$$

where $\Sigma_{1R}(p^2) = \Sigma_1 + \text{counterterm} = \Sigma_{1(0)} + \delta m^2 + \delta Z p^2$

$$\implies G_{2(0)} = \frac{\mathrm{i}Z}{p^2 - m_{ph}^2 - \Sigma_{1(0)} + \delta m^2 + \delta Z p^2} = \frac{\mathrm{i}Z}{p^2 - m_{ph}^2 - \Sigma_{1R} + \mathrm{i}\epsilon} \tag{56}$$

where $\Sigma_{1R}(m_{ph}^2) = 0$ (**On-shell renormalization condition**).

The residue of $G_{2(0)}$ is $R_{(0)}$ around $p^2 = m_{ph}^2$

$$\implies G_{2(0)} = \frac{\mathrm{i}R_{(0)}}{p^2 - m_{ph}^2} + \text{finite} \quad \text{as } p^2 \to m_{ph}^2 \tag{57}$$

In bare theory: $G_{2(0)} = \frac{\mathrm{i}R_{(0)}}{p^2 - m_{ph}^2}$ + finite, residue $= 0 \le R_{(0)} < 1$. In interaction theory: $G(p) = \frac{\mathrm{i}}{p^2 - m_{ph}^2}$, residue $= 1$. For renormalized propagator $G_2 = \frac{\mathrm{i}}{p^2 - m_{ph}^2 - \Sigma_{1R}}$. There are two on-shell requirements. The first one fixes the pole position,

$$\Sigma_{1R}(m_{ph}^2) = 0, \tag{58}$$

and the second one fixes the residue of the pole to be one. To see the second condition explicitly, expand Σ_{1R} around $p^2 = m_{ph}^2$:

$$\Sigma_{1R}(p^2) = \Sigma_{1R}(m_{ph}^2) + (p^2 - m_{ph}^2)\Sigma'_{1R}(m_{ph}^2) + O((p^2 - m_{ph}^2)^2) \tag{59}$$
$$= (p^2 - m_{ph}^2)\Sigma'_{1R}(m_{ph}^2) + O((p^2 - m_{ph}^2)^2). \tag{60}$$

Hence, near the pole,

$$G_2(p^2) \simeq \frac{\mathrm{i}}{(p^2 - m_{ph}^2)\left[1 - \Sigma'_{1R}(m_{ph}^2)\right]}. \tag{61}$$

If we choose the renormalized field so that the pole residue is one, then the second on-shell condition is

$$\Sigma'_{1R}(m_{ph}^2) = 0. \tag{62}$$

Using $G_{2(0)} = ZG_2$, we have

$$G_{2(0)}(p^2 \to m_{ph}^2) = ZG_2(p^2 \to m_{ph}^2) = \frac{\mathrm{i}Z}{p^2 - m_{ph}^2} \tag{63}$$

So if we require the residue $= 1$, then

$$Z = R_{(0)} \tag{64}$$

Thus we can require Z defined by $\phi = Z^{-1/2}\phi_0$ satisfies

$$Z = Z \times \text{finite constant} \tag{65}$$

For bare propagator, $\Sigma_{1(0)}(p^2) \simeq \Sigma_{1(0)}(m_{ph}^2) + (p^2 - m_{ph}^2)\frac{\partial}{\partial p^2}\Sigma_{1(0)}(m_{ph}^2)$

$$G_{2(0)} = \frac{\mathrm{i}}{p^2 - m_0^2 - \Sigma_{1(0)}} \simeq \frac{\mathrm{i}}{(p^2 - m_{ph}^2)(1 - \Sigma'_{1(0)})} \tag{66}$$

$$Z = \frac{1}{1 - \Sigma'_{1(0)}} = R_{(0)} \tag{67}$$

For bare ϕ^3: $\mathcal{L} = \frac{1}{2}(\partial\phi_0)^2 - \frac{1}{2}m_0^2\phi_0^2 - \frac{g_0}{3!}\phi_0^3$ def: field strength renormalization: $\phi = Z^{-1/2}\phi_0$

$$\implies \mathcal{L} = \frac{1}{2}Z(\partial\phi)^2 - \frac{1}{2}Zm_0^2\phi^2 - \frac{g_0}{3!}Z^{3/2}\phi^3 \tag{68}$$
$$= \frac{1}{2}(1 - \delta Z)(\partial\phi)^2 - \frac{1}{2}(m_{ph}^2 + \delta m^2)\phi^2 - \frac{1}{3!}(g_{ph} + \delta g)\phi^3 \tag{69}$$

where we have defined: $\begin{cases} Z = 1 - \delta Z \\ Zm_0^2 = m_{ph}^2 + \delta m^2 \\ Z^{3/2}g_0 = g_{ph} + \delta g \end{cases}$

$$\implies \mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m_{ph}^2\phi^2 - \frac{g_{ph}}{3!}\phi^3 - \frac{1}{2}\delta Z(\partial\phi)^2 - \frac{1}{2}\delta m^2\phi^2 - \frac{1}{3!}\delta g\phi^3 \tag{70}$$

Arbitrariness in a renormalization graph

Consider 1-loop graph in $d = 4$, we choose $\delta Z = 0$ so the counterterm is δm^2. The important point is that the divergent part of the counterterm is fixed by the requirement of finiteness, while its finite part is a convention. Different conventions are called different renormalization schemes. They do not change the bare theory; they only change which finite parameter we decide to call the renormalized mass.

$$\Sigma_{1R}(p^2) = \Sigma_{1,fin}(p^2) + (\delta m^2 + \Sigma_{1,div}) \tag{71}$$

In the mass-shell scheme we require that the pole of the full propagator is at $p^2 = m_{ph}^2$. This is the condition

$$\Sigma_{1R}(m_{ph}^2) = 0. \tag{72}$$

Substituting $p^2 = m_{ph}^2$ into the previous line gives

$$\Longrightarrow \delta m^2 = -\Sigma_{1,div} - \Sigma_{1,fin}(p^2 = m_{ph}^2) \tag{73}$$

$$\Longrightarrow \Sigma_{1R}(p^2) = \Sigma_{1,fin}(p^2) - \Sigma_{1,fin}(p^2 = m_{ph}^2) \tag{74}$$

$$\Longrightarrow G_2 = \frac{\mathrm{i}}{p^2 - m_{ph}^2 - \Sigma_{1R}} = \frac{\mathrm{i}}{p^2 - m_{ph}^2 - \Sigma_{1,fin}(p^2) + \Sigma_{1,fin}(m_{ph}^2) + O(g^4)} \tag{75}$$

$m_0^2 = m_{ph}^2 + \delta m^2 = m_{ph}^2 - \Sigma_{1,div} - \Sigma_{1,fin}(p^2 = m_{ph}^2) + O(g^4)$

One can also compute with different finite part to δm^2. Generally, the counterterm only requires to cancel the divergent. So it can be written as

$$\delta m^2 = -\Sigma_{1,div} + \text{finite} \tag{76}$$

Thus, we can choose

$$\delta m^2 = -\Sigma_{1,div} + g^2 C \tag{77}$$

where C is a const. The self-energy is now

$$\Sigma_{1R}^{(C)}(p^2, m^2) = \Sigma_{1,fin}(p^2) + g^2 C \tag{78}$$

Here m is the mass parameter defined by this new scheme. It is not automatically equal to the physical pole mass. The pole mass is obtained only after solving

$$p^2 - m^2 - \Sigma_{1R}^{(C)}(p^2, m^2) = 0 \tag{79}$$

order by order in g.

$$\Longrightarrow G_2 = \frac{\mathrm{i}}{p^2 - m^2 - \Sigma_{1R}^{(C)}(p^2, m^2)} = \frac{\mathrm{i}}{p^2 - m^2 - \Sigma_{1,fin}(p^2) - g^2 C + O(g^4)} \tag{80}$$

$$m_0^2 = m^2 - \Sigma_{1,div} - g^2 C + O(g^4) \tag{81}$$

Thus, we have at least two ways to renormalize the theory that gives the same result if m_0^2 are the same. So obviously we have

$$m^2 = m_{ph}^2 + g^2 C - \Sigma_{1,fin}(p^2 = m_{ph}^2) + O(g^4) \tag{82}$$

This equation is the finite redefinition between two schemes. The left hand side is the scheme-dependent mass parameter, while m_{ph} is defined by the pole and is observable.

Dimensional Regularization

As another way to deal with the divergent, Dimensional regularization is more convenient. Take the dimension as a regulator d which is a continuous variable. In the end, the cut-off will be removed by taking the limit $d \to 4$. We also use 1-loop in ϕ^3 as an example. The integral is first evaluated where it converges, and the answer is then analytically continued as a function of d. Therefore the divergence no longer appears as a large cut-off Λ; it appears as a pole of a Gamma function at the physical value of d.

$$\Sigma_1(p^2, m^2, d) = \frac{\mathrm{i}g^2}{2} \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{1}{[k^2 - m^2 + \mathrm{i}\epsilon][(p+k)^2 - m^2 + \mathrm{i}\epsilon]} \tag{83}$$

The Gaussian integral is

$$\int \mathrm{d}^n x \mathrm{e}^{-x^2} = (\sqrt{\pi})^n = \pi^{n/2} \tag{84}$$

Thus, after Wick rotation, $k^0 = \mathrm{i}\omega$, we have

$$\int \mathrm{d}^d k \mathrm{e}^{k^2} = \mathrm{i} \int \mathrm{d}\omega \int \mathrm{d}^{d-1}\vec{k}\, \mathrm{e}^{-(\omega^2 + \vec{k}^2)} \tag{85}$$

$$= \mathrm{i} \Big(\int \mathrm{d}\omega \mathrm{e}^{-\omega^2} \Big) \Big(\int \mathrm{d}^{d-1}\vec{k}\, \mathrm{e}^{-\vec{k}^2} \Big) \tag{86}$$

$$= \mathrm{i}\sqrt{\pi} \cdot \pi^{\frac{d-1}{2}} = \mathrm{i}\pi^{d/2} \tag{87}$$

Notice that

$$\frac{\mathrm{i}}{m^2 - k^2 - \mathrm{i}\epsilon} = \int_0^\infty \mathrm{d}a\, \mathrm{e}^{-\mathrm{i}a(m^2 - k^2 - \mathrm{i}\epsilon)} \tag{88}$$

Thus,

$$\Sigma_1 = \frac{\mathrm{i}g^2}{2} \int \frac{\mathrm{d}^d k}{(2\pi)^d} \Big(\int_0^\infty \mathrm{d}a\, \mathrm{e}^{-\mathrm{i}a(m^2 - k^2 - \mathrm{i}\epsilon)} \Big) \Big(\int_0^\infty \mathrm{d}b\, \mathrm{e}^{-\mathrm{i}b(m^2 - (p+k)^2 - \mathrm{i}\epsilon)} \Big) \tag{89}$$

$$= \frac{\mathrm{i}g^2}{2(2\pi)^d} \int \mathrm{d}^d k \int_0^\infty \mathrm{d}a \int_0^\infty \mathrm{d}b\, \mathrm{e}^{-\mathrm{i}[(a+b)m^2 + (a+b)k^2 + bp^2 + 2bp\cdot k]} \tag{90}$$

$$= \frac{\mathrm{i}g^2}{2(2\pi)^d} \int \mathrm{d}^d k \int_0^\infty \mathrm{d}a \int_0^\infty \mathrm{d}b\, \mathrm{e}^{-\mathrm{i}[(a+b)m^2 + (a+b)(k + \frac{b}{a+b}p)^2 + bp^2(1 - \frac{b}{a+b})]} \tag{91}$$

take $k' = k + \frac{b}{a+b}p, \quad z = a+b, \quad x = \frac{a}{z} \implies a = xz, \quad b = (1-x)z$

$$\implies \mathrm{d}a\mathrm{d}b = J\mathrm{d}x\mathrm{d}z, \quad J = \begin{vmatrix} \frac{\partial a}{\partial x} & \frac{\partial a}{\partial z} \\ \frac{\partial b}{\partial x} & \frac{\partial b}{\partial z} \end{vmatrix} = \begin{vmatrix} z & x \\ -z & 1-x \end{vmatrix} = z \tag{92}$$

$$\implies \mathrm{d}a\mathrm{d}b = z\mathrm{d}x\mathrm{d}z, \quad z \in (0, \infty), \quad x = \frac{a}{a+b} \in (0,1) \tag{93}$$

$$\implies \Sigma_1 = \frac{\mathrm{i}g^2}{2(2\pi)^d} \int \mathrm{d}^d k' \int_0^1 \mathrm{d}x \int_0^\infty \mathrm{d}z \cdot z \cdot \mathrm{e}^{-\mathrm{i}zm^2 + \mathrm{i}zk'^2 + \mathrm{i}zp^2 x(1-x)} \tag{94}$$

take $k'' = z^{1/2}k'$

$$= \frac{\mathrm{i}g^2}{2(2\pi)^d} \int_0^1 \mathrm{d}x \int_0^\infty \mathrm{d}z \cdot z^{1-d/2} \mathrm{e}^{-\mathrm{i}z(m^2 - p^2 x(1-x))} \Big(\int \mathrm{d}^d k'' \mathrm{e}^{\mathrm{i}k''^2} \Big) \tag{95}$$

$$= \frac{g^2}{2(4\pi)^{d/2}} \int_0^1 \mathrm{d}x \int_0^\infty \mathrm{d}(\alpha z)\, (\alpha z)^{(2-d/2)-1} \mathrm{e}^{-\alpha z} \quad [\Gamma(t) = \int_0^\infty \mathrm{d}z \frac{z^{t-1}}{\mathrm{e}^z}] \tag{96}$$

$$= -\frac{g^2}{2(4\pi)^{d/2}} \Gamma(2 - \frac{d}{2}) \int_0^1 \mathrm{d}x\, [m^2 - p^2 x(1-x)]^{d/2-2} \tag{97}$$

We can use Feynman parameter to get the same result:

$$\Sigma_1 = \frac{\mathrm{i}g^2}{2}\int\frac{\mathrm{d}^d k}{(2\pi)^d}\frac{1}{(k^2-m^2+\mathrm{i}\epsilon)((p+k)^2-m^2+\mathrm{i}\epsilon)} \tag{98}$$
$$= \frac{\mathrm{i}g^2}{2(2\pi)^d}\int\mathrm{d}^d k\int_0^1\mathrm{d}x\frac{1}{[k^2-m^2+xp^2+2xp\cdot k]^2} \tag{99}$$
$$= \frac{\mathrm{i}g^2}{2(2\pi)^d}\int\mathrm{d}^d k\int_0^1\mathrm{d}x\frac{1}{[(k+xp)^2+p^2x(1-x)-m^2]^2} \tag{100}$$

take $k' = xp+k$, and do Wick rotation, i.e. $k^0 = \mathrm{i}k_E^0 \implies k'^2 = -k_E^2$

$$= \frac{-\mathrm{i}g^2\cdot\mathrm{i}}{2(2\pi)^d}\int_0^1\mathrm{d}x\int\mathrm{d}^d k_E\frac{1}{[-k_E^2-m^2+p^2x(1-x)]^2} \tag{101}$$
$$= \frac{-g^2}{2(2\pi)^d}\int_0^1\mathrm{d}x\int\mathrm{d}^d k_E\frac{1}{(k_E^2+\Delta)^2},\quad \Delta = m^2-p^2x(1-x) \tag{102}$$
$$= \frac{-g^2}{2(4\pi)^{d/2}}\Gamma(2-\frac{d}{2})\int_0^1\mathrm{d}x\frac{1}{(m^2-p^2x(1-x))^{2-d/2}} \tag{103}$$

Now, the divergence only in the Γ-function's poles, at $d = 4,6,8\ldots$ The advantage of dimensional regularization is:

1. Calculation procedure is unchanged.
2. Preserve Poincaré invariance and gauge symmetries.
3. It can also be useful for IR-divergence (infra-red), e.g. $\frac{\mathrm{i}g^2}{8\pi^2}\int\mathrm{d}^4k\frac{1}{(k^2+\mathrm{i}\epsilon)((p+k)^2+\mathrm{i}\epsilon)}$

Minimal subtraction

Now we can compute the renormalized self-energy from Σ_1 above, i.e.

$$\Sigma_{1R}^{(ph)} = \Sigma_1(p^2,m_{ph}^2,g,d)-\Sigma_1(m_{ph}^2,m_{ph}^2,g,d) \tag{104}$$
$$= -\frac{g^2}{2(4\pi)^{d/2}}\Gamma(2-\frac{d}{2})\int_0^1\mathrm{d}x\left\{[m_{ph}^2-p^2x(1-x)]^{d/2-2}-[m_{ph}^2(1-x+x^2)]^{d/2-2}\right\} \tag{105}$$

To expand it, set $d = 4-\epsilon$. The two elementary expansions are

$$\Gamma(z) = \frac{1}{z}-\gamma_E+O(z),\quad z\to 0, \tag{106}$$
$$A^\delta = \mathrm{e}^{\delta\ln A} = 1+\delta\ln A+O(\delta^2),\quad \delta\to 0. \tag{107}$$

Thus

$$\Gamma\left(2-\frac{d}{2}\right) = \Gamma\left(\frac{\epsilon}{2}\right) = \frac{2}{\epsilon}-\gamma_E+O(\epsilon), \tag{108}$$
$$[m_{ph}^2-p^2x(1-x)]^{\frac{d}{2}-2} = 1-\frac{\epsilon}{2}\ln[m_{ph}^2-p^2x(1-x)]+O(\epsilon^2). \tag{109}$$

Similarly,

$$[m_{ph}^2(1-x+x^2)]^{\frac{d}{2}-2} = 1 - \frac{\epsilon}{2}\ln[m_{ph}^2(1-x+x^2)] + O(\epsilon^2). \tag{110}$$

Subtracting the two terms in the curly bracket removes the order-one part before the pole multiplies it. This is why the mass-shell subtraction gives a finite answer:

$$\begin{aligned}\Sigma_{1R}^{(ph)} &= \frac{g^2}{32\pi^2}\left(\frac{2}{\epsilon}-\gamma_E\right)\frac{\epsilon}{2}\int_0^1 \mathrm{d}x \ln\frac{m_{ph}^2 - p^2x(1-x)}{m_{ph}^2(1-x+x^2)} + O(\epsilon) &(111)\\ &= \frac{g^2}{32\pi^2}\int_0^1 \mathrm{d}x \ln\frac{m_{ph}^2 - p^2x(1-x)}{m_{ph}^2(1-x+x^2)} + O(\epsilon) &(112)\end{aligned}$$

This is the mass-shell scheme result: $\Sigma_{1R}^{(ph)}(m_{ph}^2) = 0$, the pole residue is normalized to one, and the mass parameter is the physical mass m_{ph}. which is as same as the result we have done before.

Notice that $\delta m^2 = \Sigma_{1R} - \Sigma_1(p^2) = -\Sigma_1(m_{ph}^2)$

$$= \frac{g^2}{2(4\pi)^{d/2}}\Gamma(2-\frac{d}{2})\int_0^1 \mathrm{d}x\, [m_{ph}^2(1-x+x^2)]^{\frac{d}{2}-2} \tag{113}$$

when $d = 4$, $\int_0^1 \mathrm{d}x\,[\dots]^0 = 1$, $\Gamma(2-\frac{d}{2}) \simeq \frac{1}{2-d/2} - \gamma_E \implies \delta m^2 = \frac{g^2/32\pi^2}{2-d/2} \implies$ Minimal Subtraction

Thus $\Sigma_{1R} = \Sigma_1(p^2) + \delta m^2 \quad (\Sigma_{1R}(p^2 = m^2) \neq 0,$ Counterterm only cancels divergence)

$$\begin{aligned}\Sigma_{1R}^{(MS)} &= -\frac{g^2}{2(4\pi)^{d/2}}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 \mathrm{d}x\,[m^2 - p^2x(1-x)]^{\frac{d}{2}-2} + \frac{g^2}{32\pi^2}\frac{1}{2-d/2} &(114)\\ &= -\frac{g^2}{32\pi^2}\left(1+\frac{\epsilon}{2}\ln 4\pi\right)\left(\frac{2}{\epsilon}-\gamma_E\right)\int_0^1 \mathrm{d}x\left[1-\frac{\epsilon}{2}\ln\Delta(x)\right] + \frac{g^2}{32\pi^2}\frac{2}{\epsilon} + O(\epsilon) &(115)\\ &= \frac{g^2}{32\pi^2}\int_0^1 \mathrm{d}x[\ln\Delta(x) + \gamma_E - \ln 4\pi] + O(\epsilon), \quad \Delta(x) = m^2 - p^2x(1-x). &(116)\end{aligned}$$

However, there are mass dimension in logarithm which is no-physics. The reason is we need to change the dimension of g.

$\frac{g}{3!}\phi^3 \subset \mathcal{L}, \quad [\mathcal{L}] = [m]^d, \quad [\phi] = [m]^{\frac{d-2}{2}}$

$$\implies [g] = [m]^{\frac{6-d}{2}}, \quad \text{when } d = 4, [g] = [m] \tag{117}$$

Thus, we can rewrite g as

$$g \to \mu^{\frac{4-d}{2}} g = \mu^{\epsilon/2} g \tag{118}$$

where the g shows on the right hand side has dimension $[m]^1$. This can also check on $\delta m^2 \propto g^2$, in which g is rescaled. So we have

$$\begin{aligned}\Sigma_{1R} &= \Sigma_1(p^2) + \delta m^2 &(119)\\ &= -\frac{g^2\mu^\epsilon}{2(4\pi)^{d/2}}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 \mathrm{d}x\,[\Delta(x)]^{\frac{d}{2}-2} + \frac{g^2}{32\pi^2}\frac{2}{\epsilon} &(120)\\ &= -\frac{g^2}{32\pi^2}\left(\frac{2}{\epsilon} - \gamma_E + \ln 4\pi + 2\ln\mu\right)\int_0^1 \mathrm{d}x\left[1-\frac{\epsilon}{2}\ln\Delta(x)\right] + \frac{g^2}{32\pi^2}\frac{2}{\epsilon} + O(\epsilon) &(121)\\ \implies \Sigma_{1R}^{(MS)} &= \frac{g^2}{32\pi^2}\int_0^1 \mathrm{d}x\left[\ln\left(\frac{m^2 - p^2x(1-x)}{4\pi\mu^2}\right) + \gamma_E\right] &(122)\end{aligned}$$

This renormalization prescription is called **Minimal Subtraction (MS)**, where counterterms are pure poles at the physical value of d.

$d = 6$ case

Now we apply it to $d = 6$. Since the superficial degree of divergence is

$$\omega(\Sigma_1) = d - 4 = 2, \tag{123}$$

we need to differentiate three times with respect to the external momentum in order to make the integral convergent. When we integrate back, the ambiguity is a polynomial of degree two in p. Lorentz invariance removes the linear term, so the counterterm has the form

$$\delta m^2 - p^2 \delta Z \tag{124}$$

The coupling is also continued away from six dimensions by

$$g_0 = \mu^{3-d/2}(g + \delta g), \quad \text{poles at } d = 6. \tag{125}$$

And for $\Gamma(z-1)$, we have

$$\Gamma(z+1) = z\Gamma(z) \tag{126}$$

$$\implies \Gamma(z-1) = \frac{1}{z-1}\Gamma(z) = \frac{1}{z-1}\left(\frac{1}{z} - \gamma_E + O(z)\right) \tag{127}$$

$$= -(1 + z + \cdots + z^n)\left(\frac{1}{z} - \gamma_E + O(z)\right) \tag{128}$$

$$= \left(\gamma_E - \frac{1}{z} + O(z)\right) + (-1 + O(z)) + \cdots = -\frac{1}{z} + \gamma_E - 1 + O(z) \tag{129}$$

Thus, for $d = 6 - \epsilon$, we have

$$\Sigma_1 = -\frac{g^2}{2(4\pi)^{d/2}}\Gamma(2 - \frac{d}{2}) \int_0^1 \mathrm{d}x \, [m^2 - p^2 x(1-x)]^{\frac{d}{2}-2} \tag{130}$$

$$= -\frac{g^2}{128\pi^3}\mu^{6-d}(4\pi)^{\epsilon/2}\Gamma(\frac{\epsilon}{2} - 1) \int_0^1 \mathrm{d}x \, (m^2 - p^2 x(1-x))^{1-\epsilon/2} \tag{131}$$

$$= -\frac{g^2}{128\pi^3}\left(-\frac{2}{\epsilon} + \gamma_E - 1 - \ln\left(4\pi\mu^2\right)\right) \int_0^1 \mathrm{d}x \, \Delta(x) \tag{132}$$

$$- \frac{g^2}{128\pi^3}\int_0^1 \mathrm{d}x \, \Delta(x) \ln \Delta(x) + O(\epsilon), \quad \Delta(x) = m^2 - p^2 x(1-x). \tag{133}$$

$$\Sigma_1 = -\frac{g^2}{128\pi^3}\left\{-\frac{2}{\epsilon}\left(m^2 - \frac{p^2}{6}\right) + (\gamma_E - 1)\left(m^2 - \frac{p^2}{6}\right) + \int_0^1 \mathrm{d}x \, \Delta(x) \ln \frac{\Delta(x)}{4\pi\mu^2}\right\}. \tag{134}$$

Thus the counterterm will cancel the divergence. So we have

$$\Sigma_{1R} = \Sigma_1 + \delta m^2 - \delta Z p^2 \tag{135}$$

where

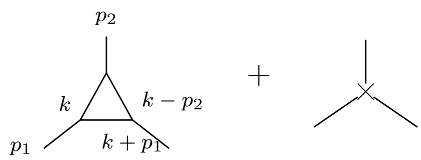
$$\delta m^2 = \frac{g^2}{64\pi^3}\frac{m^2}{d-6} + O(g^4) \tag{136}$$

$$\delta Z = \frac{g^2}{6 \cdot 64\pi^3}\frac{1}{d-6} + O(g^4) \tag{137}$$

Now we can consider δg. As a counterterm, it relates with the term in $\mathcal{L}$

$$\frac{g}{3!}\phi^3 \subset \mathcal{L} \tag{138}$$

The term δg cancels the UV divergence of the three-point vertex. So the ordinary tree vertex is corrected by the one-loop triangle, and this loop divergence is then canceled by a local three-leg counterterm:



The counterterm has no loop momentum. It is local, so in momentum space it is just a constant for this logarithmic divergence. To extract only this local UV pole, we may set the external momenta to zero after checking that the superficial divergence is logarithmic. Momentum-dependent terms are less divergent and only change the finite part. Thus, the loop is

$$\Sigma_1^{(3)} = (-\mathrm{i}g\mu^{3-d/2})^3 \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{\mathrm{i}}{k^2-m^2} \frac{\mathrm{i}}{(k+p_1)^2-m^2} \frac{\mathrm{i}}{(k-p_2)^2-m^2} \tag{139}$$

$$\xrightarrow{p\to 0} (-\mathrm{i}g\mu^{3-d/2})^3 \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{-\mathrm{i}}{(k^2-m^2)^3} = (g\mu^{3-d/2})^3 \int \frac{\mathrm{d}^d k_E}{(2\pi)^d} \frac{-\mathrm{i}}{(k_E^2+m^2)^3} \tag{140}$$

$$= (g\mu^{3-d/2})^3 \frac{-\mathrm{i}}{(4\pi)^{d/2}} \frac{\Gamma(3-d/2)}{2} m^{d-6}, \quad \text{take } d = 6-\epsilon \tag{141}$$

$$\simeq g^3\mu^{3\epsilon/2} \frac{-\mathrm{i}}{(4\pi)^3}\left(1+\frac{\epsilon}{2}\ln 4\pi\right)\frac{1}{2}\left(\frac{2}{\epsilon}-\gamma_E\right)(1-\epsilon\ln m) \tag{142}$$

$$\simeq \frac{-\mathrm{i}g^3}{2(4\pi)^3}\mu^{3\epsilon/2}\left(\frac{2}{\epsilon}-\gamma_E+\ln 4\pi-\ln m^2\right) \tag{143}$$

$$\simeq \frac{-\mathrm{i}g^3}{2(4\pi)^3}\left(1+\frac{3}{2}\epsilon\ln\mu\right)\left(\frac{2}{\epsilon}+\text{finite}\right) \tag{144}$$

$$= \frac{-\mathrm{i}g^3}{2(4\pi)^3}\left(\frac{2}{\epsilon}+3\ln\mu+\text{finite}\right) \tag{145}$$

To cancel the divergence, we introduce δg as

$$\frac{g}{3!}\phi^3 \to \frac{g}{3!}\phi^3 + \frac{\delta g}{3!}\phi^3 \tag{146}$$

where the last term should cancel the divergent $\frac{2}{\epsilon}$ and keep the order of unit mass to 1. So we have:

$$-\mathrm{i}\delta g = \frac{\mathrm{i}g^3}{2(4\pi)^3}\left(\frac{2}{\epsilon}+\ln\mu\right) = \frac{\mathrm{i}g^3}{2(4\pi)^3}\frac{2}{\epsilon}\left(1+\frac{\epsilon}{2}\ln\mu\right) \tag{147}$$

In this case, $\Sigma_1^{(3)} - \mathrm{i}\delta g = \frac{\mathrm{i}g^3}{2(4\pi)^3}(\ln\mu + \text{finite})$

$$\Longrightarrow \text{ the counterterm is } \delta g = \frac{-g^3}{64\pi^3}\frac{1}{\epsilon}\mu^{3-d/2} = \frac{g^3}{64\pi^3}\frac{1}{d-6}\mu^{3-d/2} \tag{148}$$

Massless theories

In Mass-shell scheme

$$\delta m_{ph}^2 = \frac{g^2}{2(4\pi)^{d/2}}\Gamma(2-\frac{d}{2})\int_0^1 \mathrm{d}x\,[m_{ph}^2(1-x+x^2)]^{\frac{d}{2}-2} \tag{149}$$

$$\xrightarrow{m_{ph}\to 0, d\to 4} \frac{g^2}{32\pi^2}(1+\frac{\epsilon}{2}\ln 4\pi)\left(\frac{2}{\epsilon}-\gamma_E\right)(1-\epsilon\ln m_{ph}) \tag{150}$$

$$= \frac{g^2}{32\pi^2}\left(\frac{2}{\epsilon}-\gamma_E+\ln 4\pi-\ln m_{ph}^2\right) \tag{151}$$

$$= \frac{g^2}{32\pi^2}\left(\frac{1}{2-d/2}-\ln\bigl(m_{ph}^2\bigr)+\text{finite}\right) \tag{152}$$

which fails when $m_{ph}\to 0$, i.e. IR div. We can see from MS subtraction from $d=4-\epsilon$ in above, i.e.

$$\Sigma_{1R}^{(MS)} = \frac{g^2}{32\pi^2}\left(\ln\frac{-p^2}{4\pi\mu^2}+\gamma_E-2\right) \tag{153}$$

So, here we know that the reason that the mass-shell scheme failed is because of the force is long range, so the propagator pole is not a simple pole, physically means that the particle will be never free from the other's influence.

Renormalization

The Lagrangian can be written as

$$\mathcal{L} = \mathcal{L}_0+\mathcal{L}_b+\mathcal{L}_{ct} \tag{154}$$

where $\mathcal{L}_0$ is the free Lagrangian

$$\mathcal{L}_0 = \frac{1}{2}(\partial\phi)^2-\frac{1}{2}m^2\phi^2 \tag{155}$$

$\mathcal{L}_b$ is the basic interaction

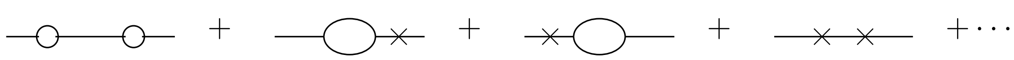
$$\mathcal{L}_b = -\frac{g}{3!}\phi^3 \tag{156}$$

$\mathcal{L}_{ct}$ is the counterterm Lagrangian

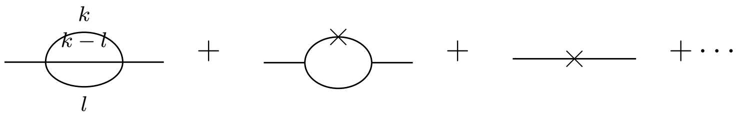
$$\mathcal{L}_{ct} = \delta Z\frac{1}{2}(\partial\phi)^2-\delta m^2\frac{1}{2}\phi^2-\frac{\delta g}{3!}\phi^3-\delta h\phi \tag{157}$$

where δh is to cancel the tadpole graphs. For order g^4, we must draw not only the genuine two-loop graphs. We must also draw the graphs where a lower-order divergent subgraph has already been replaced by its counterterm. The relevant classes are:

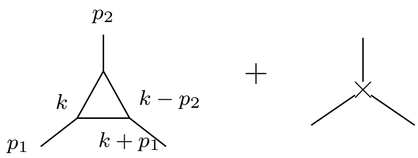
1. Two one-loop bubbles on the same propagator line:



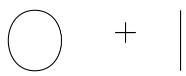
2. A two-loop self-energy graph with a one-loop subdivergence:



3. A one-loop graph with a vertex correction:



Each cross stands for a local counterterm. The point of the list is simple: every divergent subgraph needs its own subtraction. We have ignored the tadpoles, because they are local one-point divergences and can be canceled by $\delta h\phi$:

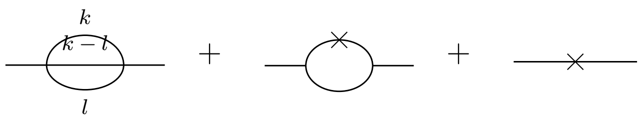


The form of divergent $\times$ logarithm is the non-local subdivergence. The form of polynomial of p is the local divergence.

Two loop self-energy in ϕ^3

Now we consider the graph (2) and (3) in $d=6$.

graph (2)



(a) (b) (c)

Graph (a) is the genuine two-loop graph. Graph (b) subtracts the one-loop subdivergence inside (a). Graph (c) is the remaining local wave-function counterterm. We need all three graphs before the final answer is finite.

$$\begin{aligned}\Sigma_{2a} = (\mathrm{i}g\mu^{3-d/2})^4 \frac{1}{2}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{\mathrm{d}^d l}{(2\pi)^d}\frac{\mathrm{i}}{k^2-m^2+\mathrm{i}\epsilon}\frac{\mathrm{i}}{l^2-m^2+\mathrm{i}\epsilon} \\ \times \frac{\mathrm{i}}{(k-l)^2-m^2+\mathrm{i}\epsilon}\frac{\mathrm{i}}{(k+p)^2-m^2+\mathrm{i}\epsilon}.\end{aligned} \tag{158}$$

For convenience, we consider the massless case, i.e.

$$\Sigma_{2a} = \frac{1}{2}g^4\mu^{12-2d}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{\mathrm{d}^d l}{(2\pi)^d}\frac{1}{(k^2+\mathrm{i}\epsilon)(l^2+\mathrm{i}\epsilon)((k-l)^2+\mathrm{i}\epsilon)((k+p)^2+\mathrm{i}\epsilon)} \tag{159}$$

After Wick rotation, we have

$$\Sigma_{2a} = -\frac{1}{2}g^4\mu^{12-2d}\int\frac{\mathrm{d}^dk_E}{(2\pi)^d}\frac{\mathrm{d}^dl_E}{(2\pi)^d}\frac{1}{[l_E^2(k_E-l_E)^2][k_E^2(k_E+p_E)^2]} \tag{160}$$

$$= -\frac{1}{2}g^4\mu^{12-2d}\int\frac{\mathrm{d}^dk_E}{(2\pi)^d}\frac{1}{k_E^2(k_E+p_E)^2}\left(\int\frac{\mathrm{d}^dl_E}{(2\pi)^d}\frac{1}{l_E^2(k_E-l_E)^2}\right) \tag{161}$$

$$= -\frac{1}{2}g^4\mu^{12-2d}\int\frac{\mathrm{d}^dk_E}{(2\pi)^d}\frac{1}{k_E^2(k_E+p_E)^2}\mathrm{i}\pi^{d/2}\frac{\Gamma(2-d/2)}{(2\pi)^d}\int_0^1\mathrm{d}x\,(-k_E^2x(1-x))^{d/2-2} \tag{162}$$

$$= -\frac{1}{2}g^4\mu^{12-2d}\int\frac{\mathrm{d}^dk_E}{(2\pi)^d}\frac{1}{k_E^2(k_E+p_E)^2}\mathrm{i}\pi^{d/2}\Gamma(2-d/2)\frac{\Gamma(\frac{d}{2}-1)^2}{\Gamma(d-2)}(-k_E^2)^{d/2-2} \tag{163}$$

$$= \frac{\mathrm{i}g^4}{2^{13}\pi^9}(16\pi^3\mu^4)^{3-d/2}\frac{\Gamma(2-d/2)\Gamma(\frac{d}{2}-1)^2}{\Gamma(d-2)}\int\mathrm{d}^dk_E\frac{1}{(-k_E^2)^{3-d/2}(k_E+p_E)^2} \tag{164}$$

For the integral, one can introduce a Feynman parameter

$$\frac{1}{A^{4-d/2}B} = \frac{\Gamma(5-d/2)}{\Gamma(4-d/2)}\int_0^1\frac{x^{3-d/2}}{[Ax+B(1-x)]^{5-d/2}} \tag{165}$$

Thus, the result is

$$\Sigma_{2a} = \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{2}\left(\frac{-p^2}{4\pi\mu^2}\right)^{d-6}\frac{\Gamma(2-d/2)\Gamma(5-d)\Gamma(\frac{d}{2}-1)^3\Gamma(d-4)}{\Gamma(d-2)\Gamma(4-d/2)\Gamma(3d/2-5)} \tag{166}$$

$$\equiv \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{2}\left(\frac{-p^2}{4\pi\mu^2}\right)^{d-6}\Gamma(2-d/2)\Gamma(5-d)K(d) \tag{167}$$

where the overall divergence is contained in $\Gamma(5-d)$. And the subdivergence is contained in $\Gamma(2-d/2)$. One can also notice that there is IR divergence since we are talking about the massless fields when $p\to 0$. It can go away if we consider the massive theory. But that will be no explicit formula for Σ_{2a}.

Now, we expand Σ_{2a} in $\epsilon = 6-d$. To keep track of the non-local part, write $L_p = \ln(-p^2/(4\pi\mu^2))$. The factors singular at $d=6$ are the two gamma functions:

$$\Sigma_{2a} = \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{2}\left(\frac{-p^2}{4\pi\mu^2}\right)^{d-6}\Gamma(2-d/2)\Gamma(5-d)K(d) \tag{168}$$

$$= \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{36}\left(\frac{1}{\epsilon}-\frac{\epsilon}{2}L_p-\frac{\epsilon^2}{4}L_p^2\right)\left(\frac{2}{\epsilon}+\gamma_E-1\right)\left(-\frac{1}{\epsilon}+\gamma_E-1\right)K_d \tag{169}$$

$$\begin{aligned} &= \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{36}\left[\frac{1}{\epsilon^2}-\frac{1}{\epsilon}\left(L_p+\frac{3}{2}(\gamma_E-1)\right)+\frac{1}{2}L_p^2\right.\\ &\qquad\left.+\frac{3}{2}(\gamma_E-1)L_p+\frac{1}{2}(\gamma_E-1)^2\right] \end{aligned} \tag{170}$$

$$= \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{36}\left[\frac{1}{(d-6)^2}+\frac{L_p+\text{Const.}}{d-6}+\frac{1}{2}L_p^2+\text{Const.}\,L_p+\text{Const.}+O(d-6)\right]. \tag{171}$$

The term proportional to $L_p/(d-6)$ is important: it is a pole multiplying a non-polynomial function of the external momentum. A local counterterm cannot cancel such a term directly. Therefore it must be canceled by the lower-order counterterm graph (b), which subtracts the divergent one-loop subgraph before the remaining overall divergence is removed. Notice for $m = 0$, the counterterm only contains $\delta_2 Z = \frac{g^2}{6\cdot 64\pi^3}\frac{1}{d-6} + O(g^4)$.

$$\Sigma_{2b} = (-\mathrm{i}g\mu^{3-d/2})^2 \int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{\mathrm{i}}{k^2-m^2}\frac{\mathrm{i}}{k^2-m^2}\frac{\mathrm{i}}{(k+p)^2-m^2}\delta_1 Z\cdot k^2 \tag{172}$$

$$= \mathrm{i}g^2\mu^{6-d}\delta_1 Z\int\frac{\mathrm{d}^d k_E}{(2\pi)^d}\frac{k_E^2}{(k_E^2)^2(k_E+p)^2} \tag{173}$$

$$= \mathrm{i}g^2\mu^{6-d}\delta_1 Z\frac{\mathrm{i}\pi^{d/2}}{(2\pi)^d}\Gamma(2-d/2)\frac{\Gamma(\frac{d}{2}-1)^2}{\Gamma(d-2)}(p^2)^{d/2-2} \tag{174}$$

$$= \left(\frac{g^2}{64\pi^3}\right)\left(\frac{p^2}{6}\right)\frac{\Gamma(2-d/2)\Gamma^2(\frac{d}{2}-1)}{(d-6)\Gamma(d-2)}\left(\frac{-p^2}{4\pi\mu^2}\right)^{d/2-3} \tag{175}$$

$$= \left(\frac{g^2}{64\pi^3}\right)^2\left(\frac{p^2}{36}\right)\left(-\frac{2}{\epsilon}+\gamma_E-1\right)\frac{1}{\epsilon}\left(1-\frac{\epsilon}{2}L_p+\frac{\epsilon^2}{8}L_p^2\right) \tag{176}$$

$$= \left(\frac{g^2}{64\pi^3}\right)^2\left(\frac{p^2}{36}\right)\left[-\frac{2}{\epsilon^2}+\frac{\gamma_E-1+L_p}{\epsilon}-\frac{1}{2}(\gamma_E-1)L_p-\frac{1}{4}L_p^2\right] \tag{177}$$

$$= \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{36}\left[-\frac{2}{(d-6)^2}+\frac{-L_p+\text{const.}}{d-6}-\frac{1}{4}L_p^2+\text{const.}\,L_p+O(d-6)\right]. \tag{178}$$

Thus, we have

$$\Sigma_{2a}+\Sigma_{2b} = \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{36}\left[\frac{-1}{(d-6)^2}+\frac{1}{d-6}\text{Const.}+\frac{1}{4}\ln^2\frac{-p^2}{4\pi\mu^2}+\text{Const.}\ln\frac{-p^2}{4\pi\mu^2}+\text{const.}+O(d-6)\right] \tag{179}$$

Now, we can see the non-local divergence has been canceled

$$\frac{1}{d-6}\left(\ln\frac{-p^2}{4\pi\mu^2}+\text{Const.}\right)\to\frac{1}{d-6}\text{const.} \tag{180}$$

The non-local subdivergence is, from its name, non-locally, which means the divergent is not because of the superposition of two space-time point on one-propagator, like the usual divergent. It has to be cancelled, otherwise we have to add extra terms onto the Lagrangian in this form

$$\phi(x)\ln(\Box)\phi(x) \tag{181}$$

for the subdivergence $\frac{1}{\epsilon}\ln(p^2)$, which is unacceptable. Since it has been cancelled, the overall divergent part left can be easily cancelled by (c), the renormalization of wave function on the counterterm is then

$$\delta_2 Z = \left(\frac{g^2}{64\pi^3}\right)^2\frac{1}{36}\left(-\frac{1}{d-6}+\frac{\text{Const.}}{d-6}\right) \tag{182}$$

Thus, the renormalized result is

$$\Sigma_{2R}^{(MS)} = \Sigma_{2a}+\Sigma_{2b}+\Sigma_{2c} = \Sigma_{2a}+\Sigma_{2b}+(-p^2\delta_2 Z) \tag{183}$$

$$= \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{36}\left(\frac{1}{4}\ln^2\frac{-p^2}{4\pi\mu^2}+\text{Const.}\ln\frac{-p^2}{4\pi\mu^2}+\text{const.}\right) \tag{184}$$

Differentiation with respect to external momenta

Now, we want to show that the counterterm of 1-PI graph is a polynomial in its external momenta with degree equal to the degree of divergence.

First, the degree of divergence for Σ_{2a} is $d-4$. To make it be negative in $d=6$, we differentiate Σ_{2a} three times.

$$\Sigma_{2a}(p) \sim \int \mathrm{d}^d k_E \mathrm{d}^d l_E \frac{1}{[l_E^2(k_E-l_E)^2][k_E^2(k_E+p_E)^2]} \tag{185}$$

$$\Longrightarrow \frac{\partial^3 \Sigma_{2a}}{\partial p^\mu \partial p^\nu \partial p^\rho} \sim \int \mathrm{d}^d k \underbrace{\left(\int \mathrm{d}^d l \frac{1}{l^2(k-l)^2}\right)}_{\text{subgraph}} \frac{\partial^3}{\partial p^3}\left(\frac{1}{k^2(k+p)^2}\right) \tag{186}$$

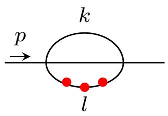
Notice that

$$\frac{\partial}{\partial p^\mu}\frac{1}{p^2-m^2} = -\frac{1}{(p^2-m^2)^2}\frac{\partial}{\partial p^\mu}(p^2-m^2) = -\frac{2p_\mu}{(p^2-m^2)^2}. \tag{187}$$

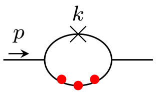
Thus differentiating a propagator produces one more propagator power and a numerator proportional to the momentum through that line. In graph language we denote this by adding a marked insertion on the differentiated line:

$$\frac{\partial}{\partial p^\mu}(\text{———}) \quad \longrightarrow \quad \text{———}\cdot \tag{188}$$

The three red dots below mean that the external-momentum dependent propagator has been differentiated three times. They are not new interaction vertices; they only remind us where the powers of momentum in the numerator come from. Hence $\frac{\partial^3 \Sigma_{2a}}{\partial p^\mu \partial p^\nu \partial p^\rho}$ should then be

(a''') 

and $\frac{\partial^3 \Sigma_{2b}}{\partial p^\mu \partial p^\nu \partial p^\rho}$ should be

(b''') 

Thus, $(a''')+(b''')$ is finite. So the overall counterterm is quadratic in P, i.e.

$$A(d)p^2 + B^\mu(d)p_\mu + C(d) \tag{189}$$

the Lorentz invariance requires $B^\mu = 0$

$$\Longrightarrow A(d)p^2 + C(d) \tag{190}$$

Now, we prove it. Consider $l \gg k \to \infty$ ($l \sim k \to \infty$ is overall divergence cancelled by differentiation; k fixed, $l \to \infty$ cancelled by (b'''))

$$(a''') \sim \int^\infty \mathrm{d}k \frac{1}{k^4}\int_{l\gg k} \mathrm{d}l \cdot l \sim l^2 \to \text{quadratic diverges.} \tag{191}$$

The remaining question is the region $l \gg k$. In this region the loop momentum l is much larger than the momentum k entering the subgraph, so the inner loop can be expanded in powers of k/l. Let

$$I(k) = \int \mathrm{d}^6 l \, \frac{1}{l^2 (l+k)^2}. \tag{192}$$

Here we keep $d = 6$, because this is the case where the two-loop graph has the subdivergence we want to subtract. The useful expansion is

$$(l+k)^2 = l^2 + 2l \cdot k + k^2 = l^2 \left(1 + \frac{2l \cdot k + k^2}{l^2} \right), \tag{193}$$

$$\frac{1}{(l+k)^2} = \frac{1}{l^2} \left[1 - \frac{2l \cdot k + k^2}{l^2} + \left(\frac{2l \cdot k + k^2}{l^2} \right)^2 + \cdots \right]. \tag{194}$$

This is the same as the Taylor expansion

$$I(k) = I(0) + k^\mu \left. \frac{\partial I}{\partial k^\mu} \right|_{k=0} + \frac{1}{2} k^\mu k^\nu \left. \frac{\partial^2 I}{\partial k^\mu \partial k^\nu} \right|_{k=0} + \cdots . \tag{195}$$

Now look term by term. In six dimensions $\mathrm{d}^6 l = \Omega_5 l^5 \mathrm{d}l$, and angular symmetry gives $\langle (l \cdot k)^2 \rangle = l^2 k^2 / 6$. Therefore

$$k^0 : \quad \int \mathrm{d}^6 l \, \frac{1}{l^4} \sim \Omega_5 \int^\Lambda \mathrm{d}l \, l \sim \Lambda^2, \qquad \text{quadratic divergence,} \tag{196}$$

$$k^1 : \quad - \int \mathrm{d}^6 l \, \frac{2l \cdot k}{l^6} = -2k_\mu \int \mathrm{d}^6 l \, \frac{l^\mu}{l^6} = 0, \qquad \text{odd integrand,} \tag{197}$$

$$\begin{aligned} k^2 : \quad & \int \mathrm{d}^6 l \left(-\frac{k^2}{l^6} + \frac{4(l \cdot k)^2}{l^8} \right) \sim \Omega_5 \int^\Lambda \mathrm{d}l \, l^5 \left(-\frac{k^2}{l^6} + \frac{4}{l^8} \frac{l^2 k^2}{6} \right) \\ & = -\frac{\Omega_5 k^2}{3} \int^\Lambda \frac{\mathrm{d}l}{l} \sim -\frac{\Omega_5 k^2}{3} \ln \Lambda, \qquad \text{logarithmic divergence.} \end{aligned} \tag{198}$$

So the counterterm in graph (b) removes exactly these local divergent pieces. The k^0 term is a constant, and the k^2 term is quadratic in the external momentum. After this subtraction, only the dependence on the matching scale k remains:

$$k^0 : \quad \int_1^\Lambda \mathrm{d}l \, l - \text{subtraction} = O(k^2), \tag{199}$$

$$k^2 : \quad \int_k^\Lambda \frac{\mathrm{d}l}{l} - \ln \Lambda = -\ln k = O(\ln k). \tag{200}$$

Terms of order k^3 and higher are already finite in the inner loop. They do not add new ultraviolet divergences. Thus the worst remaining behavior in the outer integral is

$$\int \mathrm{d}k \frac{1}{k^4} (k^2 \ln k) = \int \mathrm{d}k \frac{\ln k}{k^2} \tag{201}$$

which is finite.

Composite Operators

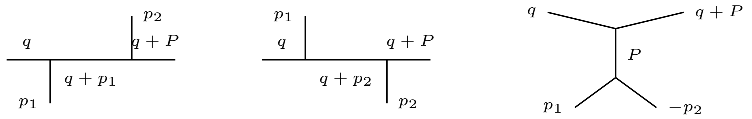
Goal: Renormalize products of operators, e.g., $\phi(x)\phi(y)$. When the separation $x-y$ becomes small, the product can be represented by local operators at one point. Equivalently, after Fourier transform, this is an expansion in the limit where the momentum flowing through the pair is much larger than the other external momenta:

$$\phi(x)\phi(y) \sim C_1(x-y)\mathbb{1} + C_{\phi^2}(x-y)[\phi^2(y)] + C_\phi(x-y)[\phi(y)] + \dots \tag{202}$$

where $[A(x)]$ denotes the renormalized operator, while $A(x)$ without brackets is the unrenormalized composite operator. The functions C_i are c-numbers and are called Wilson coefficients.

operator product expansion

Consider the graphs for four-point function in ϕ^3-theory. In the labels of the figures, we write $P = p_1 + p_2$.



(a) hard (b) hard (c) soft

where $q^\mu \to \infty$, and p_1, p_2 fixed. The words "hard" and "soft" only tell us which part carries the large momentum. In (a) and (b), all three propagators on the top line carry momenta of order q. In (c), only the two upper propagators are hard; the lower line with momentum $P = p_1 + p_2$ is soft and will become a local operator. The expansion used below is simply

$$\frac{1}{(q+p)^2 - m^2} = \frac{1}{q^2}\left[1 - \frac{2q\cdot p + p^2 - m^2}{q^2} + O\left(\frac{1}{q^2}\right)\right]. \tag{203}$$

For the soft denominators it is useful to write

$$P = p_1 + p_2, \qquad D_1 = p_1^2 - m^2, \qquad D_2 = p_2^2 - m^2, \qquad D_P = P^2 - m^2.$$

Only the propagators carrying q are expanded; D_1, D_2, D_P are kept fixed.

$$(a) = (-\mathrm{i}g)^2 \frac{\mathrm{i}}{q^2-m^2}\frac{\mathrm{i}}{D_1}\frac{\mathrm{i}}{(q+p_1)^2-m^2}\frac{\mathrm{i}}{(q+P)^2-m^2}\frac{\mathrm{i}}{D_2} \tag{204}$$

$$\xrightarrow{q\to\infty} \mathrm{i}g^2\left(\frac{\mathrm{i}}{D_1}\frac{\mathrm{i}}{D_2}\right)\frac{1}{(q^2)^3} + O\left(\frac{1}{q^7}\right), \tag{205}$$

$$(b) \xrightarrow{q\to\infty} \mathrm{i}g^2\left(\frac{\mathrm{i}}{D_1}\frac{\mathrm{i}}{D_2}\right)\frac{1}{(q^2)^3} + O\left(\frac{1}{q^7}\right), \tag{206}$$

$$\implies (a) + (b) \simeq 2\mathrm{i}g^2\left(\frac{\mathrm{i}}{D_1}\frac{\mathrm{i}}{D_2}\right)\frac{1}{(q^2)^3}, \tag{207}$$

$$(c) = (-\mathrm{i}g)^2 \frac{\mathrm{i}}{q^2-m^2}\frac{\mathrm{i}}{(q+P)^2-m^2}\frac{\mathrm{i}}{D_P}\frac{\mathrm{i}}{D_1}\frac{\mathrm{i}}{D_2}. \tag{208}$$

$$(c) = \frac{-\mathrm{i}g^2}{(q^2)^2}\left[1 - \frac{2q\cdot P}{q^2} + \frac{2m^2 - P^2}{q^2} + \frac{4(q\cdot P)^2}{(q^2)^2} + \cdots\right]$$
$$\times \frac{1}{(P^2-m^2)(p_1^2-m^2)(p_2^2-m^2)}, \qquad P = p_1 + p_2. \tag{209}$$

Now, we will show that it can be expanded like an operator-product. For (a) + (b),

$$(a)+(b) \simeq \underbrace{\frac{2\mathrm{i}g^2}{(q^2)^3}}_{\text{hard coefficient}} \underbrace{\left(\frac{\mathrm{i}}{p_1^2-m^2}\frac{\mathrm{i}}{p_2^2-m^2}\right)}_{\text{soft matrix element}}$$

$$= C_{\phi^2}(q)\ \langle 0|\,\mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(0)\}\,|0\rangle\,. \tag{210}$$

$\Longrightarrow$ For this four-point correlation function, after the operator $\frac{1}{2}\phi^2(0)$ is fixed at the origin, only two spacetime points are left free: x and y (or, in momentum space, p_1 and p_2). Hence

$$\langle 0|\,\mathrm{T}\{\phi(x)\phi(y)(\frac{1}{2}\phi(0)\phi(0))\}\,|0\rangle \tag{211}$$

$$= \langle 0|\,\mathrm{T}\{\phi(x)\phi(y)(\frac{1}{2}\phi(0)\phi(0))\}\,|0\rangle_c + \langle 0|\,\mathrm{T}\{\phi(x)\phi(0)\}\,|0\rangle\,\langle 0|\,\mathrm{T}\{\phi(y)\phi(0)\}\,|0\rangle \tag{212}$$

$$= \frac{1}{2}D(x)D(y) + \frac{1}{2}D(y)D(x) = D(x)D(y) \tag{213}$$

In momentum space:

$$\int \mathrm{d}^4x\mathrm{d}^4y\,\mathrm{e}^{-\mathrm{i}p_1\cdot x}\mathrm{e}^{-\mathrm{i}p_2\cdot y}D(x)D(y) = \frac{\mathrm{i}}{p_1^2-m^2}\frac{\mathrm{i}}{p_2^2-m^2} \tag{214}$$

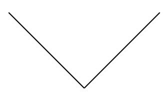
$$\Longrightarrow \int \mathrm{d}^4x\mathrm{d}^4y\,\mathrm{e}^{-\mathrm{i}p_1\cdot x}\mathrm{e}^{-\mathrm{i}p_2\cdot y}\ \langle 0|\,\mathrm{T}\{\phi(x)\phi(y)\frac{1}{2}\phi^2(0)\}\,|0\rangle \tag{215}$$

$$= \langle 0|\,\mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(0)\}\,|0\rangle \tag{216}$$

Here $\phi(0)$ is not Fourier transformed, because it is the local operator inserted at the origin. To see why the factor $\frac{1}{2}\phi^2(0)$ gives this rule, use the functional integral definition:

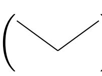
$$\langle 0|\,\mathrm{T}\{\phi(x)\phi(y)\frac{1}{2}\phi^2(0)\}\,|0\rangle = N\int [\mathrm{d}A]A(x)A(y)\frac{1}{2}A^2(0)\mathrm{e}^{\mathrm{i}S} \tag{217}$$

Here ϕ denotes the quantum operator, while A is the classical field inside the path integral. At lowest order, the insertion of $\frac{1}{2}\phi^2(0)$ therefore gives one local black-dot vertex with two scalar lines:



This insertion $\frac{1}{2}\phi^2(0)$ is our first example of a composite operator. Therefore

$$(a)+(b) \sim \frac{2\mathrm{i}g^2}{(q^2)^3}\ \langle 0|\,\mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(0)\}\,|0\rangle \tag{218}$$

$$= \frac{2\mathrm{i}g^2}{(q^2)^3}\Big($$  $$\Big) \tag{219}$$

For (c), firstly, notice

$$\langle 0|\,\mathrm{T}\{\phi(x)\phi(y)\phi(z)\}\,|0\rangle_{\text{interaction vacuum}} = \langle 0|\,\mathrm{T}\{\phi(x)\phi(y)\phi(z)\int \mathrm{d}^4w\frac{-\mathrm{i}g}{3!}\phi^3(w)\}\,|0\rangle_{\text{free vacuum}} \tag{220}$$

$$= -\mathrm{i}g\int \mathrm{d}^4w\, D(x-w)D(y-w)D(z-w) \tag{221}$$

Use

$$D(u-v)=\int\frac{\mathrm{d}^4k}{(2\pi)^4}\mathrm{e}^{\mathrm{i}k\cdot(u-v)}\frac{\mathrm{i}}{k^2-m^2}.$$

In momentum space, with $P=p_1+p_2$,

$$\begin{aligned}
\langle 0|\,\mathrm{T}\{\tilde\phi(p_1)\tilde\phi(p_2)\phi(x)\}\,|0\rangle &= \int \mathrm{d}^4y\mathrm{d}^4z\,\mathrm{e}^{-\mathrm{i}p_1\cdot y-\mathrm{i}p_2\cdot z}\,\langle 0|\,\mathrm{T}\{\phi(x)\phi(y)\phi(z)\}\,|0\rangle && (222)\\
&= -\mathrm{i}g\left(\frac{\mathrm{i}}{p_1^2-m^2}\frac{\mathrm{i}}{p_2^2-m^2}\right)\int \mathrm{d}^4w\,\mathrm{e}^{-\mathrm{i}P\cdot w}D(x-w) && (223)\\
&= -\mathrm{i}g\left(\frac{\mathrm{i}}{p_1^2-m^2}\frac{\mathrm{i}}{p_2^2-m^2}\right)\int\frac{\mathrm{d}^4k}{(2\pi)^4}\frac{\mathrm{i}\,\mathrm{e}^{\mathrm{i}k\cdot x}}{k^2-m^2}\int \mathrm{d}^4w\,\mathrm{e}^{-\mathrm{i}(P+k)\cdot w} && (224)\\
&= -\mathrm{i}g\left(\frac{\mathrm{i}}{p_1^2-m^2}\frac{\mathrm{i}}{p_2^2-m^2}\right)\frac{\mathrm{i}}{P^2-m^2}\mathrm{e}^{-\mathrm{i}P\cdot x}. && (225)
\end{aligned}$$

Thus, we can take $x=0$ such that the $\mathrm{e}^{-\mathrm{i}(p_1+p_2)\cdot x}=1$. And also, if we only consider the propagator, the $-\mathrm{i}g$ belongs to the vertex, so it's not considered here. When this matrix element is written as a local operator at $x=0$, the momentum P is represented by derivatives acting on the local field:

$$P_\mu\to\mathrm{i}\partial_\mu,\qquad P^2\to-\Box.$$

$$(c)\sim\frac{\mathrm{i}g^2}{(q^2)^2}\left[1-\frac{2q\cdot P}{q^2}+\frac{2m^2-P^2}{q^2}+\frac{4(q\cdot P)^2}{(q^2)^2}+\cdots\right]\left[\frac{-\mathrm{i}}{(P^2-m^2)(p_1^2-m^2)(p_2^2-m^2)}\right] \tag{226}$$

$$=\frac{\mathrm{i}g^2}{(q^2)^2}\left[1-\frac{2\mathrm{i}q^\mu\partial_\mu}{q^2}+\frac{2m^2+\Box}{q^2}-\frac{4q^\mu q^\nu\partial_\mu\partial_\nu}{(q^2)^2}+\cdots\right]\langle 0|\,\mathrm{T}\{\tilde\phi(p_1)\tilde\phi(p_2)\phi(x)\}\,|0\rangle\Big|_{x=0} \tag{227}$$

$$=\frac{\mathrm{i}g^2}{(q^2)^2}\sum\text{coefficients}\times\text{derivatives}\left(\bigvee\right) \tag{228}$$

Thus, we have

$$\begin{aligned}
&\langle 0|\,\mathrm{T}\{\tilde\phi(q)\phi(0)\tilde\phi(p_1)\tilde\phi(p_2)\}\,|0\rangle\\
&\quad\sim\sum_i C_i(q)\,\langle 0|\,\mathrm{T}\{O_i(0)\tilde\phi(p_1)\tilde\phi(p_2)\}\,|0\rangle\,.
\end{aligned}\tag{229}$$

The sum is over a set of local operators O_i. Each of these is either the elementary field ϕ, one of its derivatives, or a composite operator, such as ϕ^2. e.g.

$$\begin{aligned}
&i=1\quad O_1=\frac{1}{2}\phi^2\quad((a),(b)) && (230)\\
&i=2\quad O_2=\phi\quad((c)\text{ leading order}) && (231)\\
&i=3\quad O_3=\partial_\mu\phi\text{ or }\Box\phi\quad((c)\text{ higher order}) && (232)
\end{aligned}$$

The functions $C_i(q)$ are the Wilson coefficients. They contain the short-distance, high-momentum part of the graph. The matrix elements of O_i contain the long-distance part.

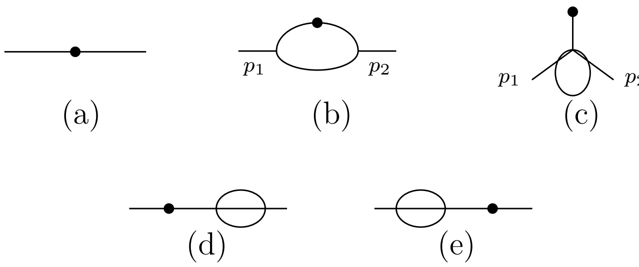
Renormalization of composite operators

Renormalization of ϕ^2

Consider ϕ^3 theory in 6-dim.

$$\langle 0|\,\mathrm{T}\{\phi(x)\phi(y)\frac{1}{2}\phi^2(z)\}\,|0\rangle \tag{233}$$

The connected graphs up to order g^2 are shown below. In these pictures, the black dot is the insertion of the composite operator $\frac{1}{2}\phi^2$. A crossed mark, when it appears later, means an ordinary counterterm from the Lagrangian.



To work on momentum space, we have

$$\begin{aligned} G &= \langle 0|\,\mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(z)\}\,|0\rangle \\ &= \int \mathrm{d}^d x\,\mathrm{d}^d y\,\mathrm{e}^{\mathrm{i}(p_1\cdot x+p_2\cdot y)}\,\langle 0|\,\mathrm{T}\{\phi(x)\phi(y)\frac{1}{2}\phi^2(z)\}\,|0\rangle\,. \end{aligned} \tag{234}$$

The lowest order graph, i.e., (a) is

$$S_i = \frac{\mathrm{i}}{p_i^2 - m^2 + \mathrm{i}\epsilon}, \qquad S_{12} = S_1 S_2.$$

$$G_a = S_{12}. \tag{235}$$

For the order g^2 graphs, the large-k behavior is

- (b) $\sim \int \mathrm{d}^6 k \frac{1}{k^2}\frac{1}{k^2}\frac{1}{k^2} \sim \int \mathrm{d}k \frac{1}{k} \implies$ logarithmically divergent.
- (c), (d), (e) $\sim \int \mathrm{d}^6 k \frac{1}{k^4} \sim \int \mathrm{d}k\, k \implies$ quadratically divergent.

Graphs (d) and (e) are ordinary self-energy corrections on the external lines. Their divergences are cancelled by the usual wave-function and mass counterterms. They do not give a new counterterm for $[\phi^2]$.

Graphs (b) and (c) are different. Their short-distance divergence sits at the composite-operator insertion itself, so the bare operator ϕ^2 is not enough. We must add local counterterms to define the finite operator $[\phi^2]$.

$$G_b = S_{12}(-\mathrm{i}g)^2 \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{\mathrm{i}}{k^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{(p_1-k)^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{(p_2+k)^2 - m^2 + \mathrm{i}\epsilon}. \tag{236}$$

After the continuation $g \to g\mu^{(6-d)/2}$, the loop part is

$$\begin{aligned} G_b &= S_{12}\left(\frac{\mathrm{i}g^2\mu^{6-d}}{(2\pi)^d}\right) \\ &\quad \times \int \mathrm{d}^d k\, \frac{1}{(k^2-m^2)[(p_1-k)^2-m^2][(p_2+k)^2-m^2]}. \end{aligned} \tag{237}$$

$$\frac{1}{(k^2-m^2)[(p_1-k)^2-m^2][(p_2+k)^2-m^2]}$$
$$= 2\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left[(1-x-y)(k^2-m^2)+x((p_1-k)^2-m^2)+y((p_2+k)^2-m^2)\right]^{-3} \tag{238}$$
$$= 2\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left[k^2-m^2-2(xp_1-yp_2)\cdot k+xp_1^2+yp_2^2\right]^{-3}. \tag{239}$$

Completing the square with $k'=k-xp_1+yp_2$ and $P=p_1+p_2$ gives

$$2\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left[(k-xp_1+yp_2)^2-(xp_1-yp_2)^2+xp_1^2+yp_2^2-m^2\right]^{-3}$$
$$= 2\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left[k'^2-\Delta_b(x,y)\right]^{-3}, \tag{240}$$
$$\Delta_b(x,y) = m^2-p_1^2x(1-x-y)-p_2^2y(1-x-y)-P^2xy. \tag{241}$$

After Wick rotation, the momentum integral is

$$\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \int \mathrm{d}^dk_E \, \frac{2}{(k_E^2+\Delta_b)^3}$$
$$= \int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \, \frac{(2\pi)^d}{(4\pi)^{d/2}}\Gamma\left(3-\frac{d}{2}\right)\Delta_b^{d/2-3}. \tag{242}$$

Therefore

$$G_b = S_{12}\frac{g^2}{64\pi^3}\Gamma\left(3-\frac{d}{2}\right)$$
$$\times \int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left(\frac{\Delta_b(x,y)}{4\pi\mu^2}\right)^{d/2-3}. \tag{243}$$

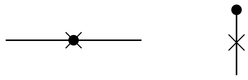
And for (c), use the same P and set

$$D_{12P} = (p_1^2-m^2)(p_2^2-m^2)(P^2-m^2).$$

Then

$$G_c = (-\mathrm{i}g\mu^{\frac{6-d}{2}})^2\frac{-\mathrm{i}}{D_{12P}}\frac{1}{2!}\int \frac{\mathrm{d}^dk}{(2\pi)^d}\frac{\mathrm{i}}{k^2-m^2}\frac{\mathrm{i}}{(k+P)^2-m^2} \tag{244}$$
$$= \frac{-\mathrm{i}g^2\mu^{6-d}}{D_{12P}}\int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^dk}{(2\pi)^d}\left[(k+xP)^2-\Delta_c(x)\right]^{-2},$$
$$\Delta_c(x) = m^2-P^2x(1-x) \tag{245}$$
$$= \frac{\frac{1}{2}g^2\mu^{6-d}}{D_{12P}}\int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^dk_E}{(2\pi)^d}\frac{1}{(k_E^2+\Delta_c(x))^2} \tag{246}$$
$$= \frac{\frac{1}{2}g^2(\mu^2)^{3-d/2}}{D_{12P}}\frac{1}{(4\pi)^{d/2}}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 \mathrm{d}x \, \Delta_c(x)^{d/2-2} \tag{247}$$
$$= \frac{g^2}{D_{12P}}\frac{1}{128\pi^3}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 \mathrm{d}x \, \frac{\Delta_c(x)^{d/2-2}}{(4\pi\mu^2)^{d/2-3}} \tag{248}$$
$$\implies G_c = \frac{-g\mu^{3-d/2}}{D_{12P}}\frac{-g\mu^{d/2-3}}{128\pi^3}\Gamma\left(2-\frac{d}{2}\right)$$
$$\times \int_0^1 \mathrm{d}x \, \frac{\Delta_c(x)^{d/2-2}}{(4\pi\mu^2)^{d/2-3}} \tag{249}$$

G_b and G_c both diverge, so the bare composite operator ϕ^2 does not define finite Green functions by itself. For the operator-product expansion we need a local operator with the same quantum numbers and a finite insertion. In the MS scheme, the required counterterms are obtained by replacing each divergent short-distance loop, i.e.,



The counterterm is: for (b):

$$
\begin{aligned}
G_b &= S_{12}\frac{g^2}{64\pi^3}\Gamma\left(3-\frac{d}{2}\right)\int_0^1 \mathrm{d}x\int_0^{1-x}\mathrm{d}y\left(\frac{\Delta_b(x,y)}{4\pi\mu^2}\right)^{d/2-3} && (250)\\
&= S_{12}\frac{g^2}{64\pi^3}\int_0^1 \mathrm{d}x\int_0^{1-x}\mathrm{d}y\left[\frac{2}{\epsilon}-\gamma_E-\ln\frac{\Delta_b(x,y)}{4\pi\mu^2}\right]+O(\epsilon) && (251)\\
&= \frac{1}{(p_1^2-m^2)(p_2^2-m^2)}\frac{g^2}{64\pi^3(d-6)}+\text{finite}. && (252)\\
\implies \text{Counterterm} &= S_{12}\frac{g^2}{64\pi^3(d-6)}. && (253)
\end{aligned}
$$

for (c):

$$
\begin{aligned}
G_c &= \frac{-g\mu^{3-d/2}}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\frac{-g\mu^{d/2-3}}{128\pi^3}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 \mathrm{d}x\,\frac{\Delta_c(x)^{d/2-2}}{(4\pi\mu^2)^{d/2-3}}, && (254)\\
\Delta_c(x) &= m^2-P^2x(1-x), \qquad P=p_1+p_2,\\
G_c &= \frac{-g\mu^{3-d/2}}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\frac{-g\mu^{d/2-3}}{128\pi^3}\left(-\frac{2}{\epsilon}\right)\int_0^1 \mathrm{d}x\,\Delta_c(x)+\text{finite} && (255)\\
&= \frac{-g\mu^{3-d/2}}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\frac{g\mu^{d/2-3}}{64\pi^3(d-6)}\left(m^2-\frac{P^2}{6}\right)+\text{finite}. && (256)
\end{aligned}
$$

$$
\implies \text{Counterterm} = \frac{-g\mu^{3-d/2}}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\left\{\frac{g\mu^{d/2-3}}{64\pi^3(d-6)}\left(m^2-\frac{P^2}{6}\right)\right\}. \qquad (257)
$$

Thus, the renormalized value at $d=6$ is

$$
\begin{aligned}
R(G_b) &= \frac{\mathrm{i}^2}{(p_1^2-m^2)(p_2^2-m^2)}\frac{g^2}{64\pi^3}\left[-\frac{\gamma_E}{2}-\int_0^1 \mathrm{d}x\int_0^{1-x}\mathrm{d}y\,\ln\frac{\Delta_b(x,y)}{4\pi\mu^2}\right], && (258)\\
R(G_c) &= \frac{-g}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\frac{-g}{128\pi^3}\left[(\gamma_E-1)\left(m^2-\frac{P^2}{6}\right)+\int_0^1 \mathrm{d}x\,\Delta_c(x)\ln\frac{\Delta_c(x)}{4\pi\mu^2}\right]. && (259)
\end{aligned}
$$

Now, in last section, we found that

$$
\begin{aligned}
(a)+(b) &= 2\mathrm{i}g^2\left(\frac{\mathrm{i}}{p_1^2-m^2}\frac{\mathrm{i}}{p_2^2-m^2}\right)\frac{1}{(q^2)^3}\\
&= \frac{2\mathrm{i}g^2}{(q^2)^3}\langle 0|\,\mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(0)\}\,|0\rangle = \frac{2\mathrm{i}g^2}{(q^2)^3}\big(\quad\big). && (260)
\end{aligned}
$$

comparing with the counterterm, we find

$$
\begin{aligned}
&-\frac{1}{(p_1^2-m^2)(p_2^2-m^2)}\frac{g^2}{64\pi^3(d-6)} \\
&\quad=\frac{g^2}{64\pi^3(d-6)}\left\langle 0\right| \mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(0)\}\left|0\right\rangle . \qquad (261)
\end{aligned}
$$

And also

$$
\begin{aligned}
(c) &= \frac{\mathrm{i}g^2}{(q^2)^2}\left[1-\frac{2q\cdot P}{q^2}+\frac{2m^2-P^2}{q^2}+\frac{4(q\cdot P)^2}{(q^2)^2}+\cdots\right]\frac{-\mathrm{i}}{D_{12P}} \qquad (262)\\
&= \frac{\mathrm{i}g^2}{(q^2)^2}\left[1-\frac{2\mathrm{i}q^\mu\partial_\mu}{q^2}+\frac{2m^2+\Box}{q^2}-\frac{4q^\mu q^\nu\partial_\mu\partial_\nu}{(q^2)^2}+\cdots\right]\left\langle 0\right| \mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\phi(x)\}\left|0\right\rangle\Big|_{x=0} \qquad (263)\\
&= \frac{\mathrm{i}g^2}{(q^2)^2}\sum \text{coefficients}\times\text{derivatives}\left(\qquad\right). \qquad (264)
\end{aligned}
$$

comparing with the counterterm, we find

$$
\begin{aligned}
&\frac{-g\mu^{3-d/2}}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\frac{g\mu^{d/2-3}}{64\pi^3(d-6)}\left(m^2-\frac{P^2}{6}\right) \\
&\quad=\frac{g\mu^{d/2-3}}{64\pi^3(d-6)}\left(m^2+\frac{1}{6}\Box\right)\left\langle 0\right| \mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\phi(0)\}\left|0\right\rangle . \qquad (265)
\end{aligned}
$$

Thus, we have ($R(G_d)$ and $R(G_e)$ have no contribution due to the divergents are cancelled by the usual counterterms)

$$
\begin{aligned}
&G_a+R(G_b)+R(G_c)+R(G_d)+R(G_e) \\
&=\left(1+\frac{g^2}{64\pi^3(d-6)}\right)\left\langle 0\right| \mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(0)\}\left|0\right\rangle \\
&\quad+\frac{g\mu^{d/2-3}}{64\pi^3(d-6)}\left\langle 0\right| \mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\left(m^2+\frac{1}{6}\Box\right)\phi(0)\}\left|0\right\rangle+O(g^4). \qquad (266)
\end{aligned}
$$

So what we are computing is a Green's function of the operator

$$
\begin{aligned}
\frac{1}{2}[\phi^2] &\equiv \left(1+\frac{g^2}{64\pi^3(d-6)}\right)\frac{1}{2}\phi^2+\frac{g\mu^{d/2-3}}{64\pi^3(d-6)}\left(m^2+\frac{1}{6}\Box\right)\phi+O(g^4) \qquad (267)\\
&= \left(1+\frac{g^2}{64\pi^3(d-6)}\right)\frac{1}{2}\phi^2+\frac{g\mu^{d/2-3}}{64\pi^3(d-6)}m^2\phi+\frac{g\mu^{d/2-3}}{6\cdot 64\pi^3(d-6)}\Box\phi. \qquad (268)
\end{aligned}
$$

Here we use $[\phi^2]$ to denote a renormalized operator. We will see later that this result can be generalized to all orders, which can be written as

$$
\frac{1}{2}[\phi^2]=Z_a\frac{1}{2}\phi^2+\mu^{d/2-3}Z_b m^2\phi+\mu^{d/2-3}Z_c\Box\phi \qquad (269)
$$

where Z's only depend on g and d. The reason that only these operators appear is the same power-counting logic used before: counterterms must be local, and an operator can mix only with operators whose dimension is not larger than its own. For ϕ^2 in six-dimensional ϕ^3 theory, the allowed lower-or-equal operators are ϕ^2, $m^2\phi$, and $\Box\phi$. We can also write the mixing as

$$
\begin{pmatrix}\frac{1}{2}[\phi^2]\\ \phi\\ \Box\phi\end{pmatrix}=\begin{pmatrix}Z_a & \mu^{d/2-3}Z_b m^2 & \mu^{d/2-3}Z_c\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}\frac{1}{2}\phi^2\\ \phi\\ \Box\phi\end{pmatrix} \qquad (270)
$$